

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3– Function Calling

Course Code:	CSA6503	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO2 – Precision (BL1, BL2)	Assessment:	Assessment Tool 2– Function calling
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models. (BL1, BL2)</i>		
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Section / Batch:	Slot A/2024	Date:	06/08/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

8. Library Management Assistant

Introduction

A **Library Management Assistant** is an AI-based system that helps users manage library books easily. It can help users search for books, check book availability, borrow books, return books, and view the list of available books. Prompt engineering can be used to make the assistant understand user requests and provide clear responses.

Objective

The main objective is to create a simple assistant that can:

- Search for books.
- Check book availability.
- Borrow books.
- Return books.
- Display available books.
- Provide simple responses to users.

Applications

1. Managing books in libraries.
2. Searching books quickly.
3. Checking book availability.
4. Tracking borrowed books.
5. Helping students find required books.

Simple Python Code

```
# Library Management Assistant
```

```
books = {  
    "python": True,  
    "java": True,  
    "c++": False,  
    "machine learning": True,  
    "data science": True  
}
```

```
print("==== Library Management Assistant =====")
```

```
while True:  
    print("\n1. Search Book")  
    print("2. Borrow Book")  
    print("3. Return Book")  
    print("4. Show Available Books")  
    print("5. Exit")
```

```
choice = input("Enter your choice: ")
```

```
if choice == "1":  
    book = input("Enter book name: ").lower()
```

```
    if book in books:  
        if books[book]:  
            print(book, "is available.")  
        else:  
            print(book, "is currently borrowed.")  
    else:  
        print("Book not found.")
```

```
elif choice == "2":  
    book = input("Enter book name: ").lower()
```

```
    if book in books:  
        if books[book]:
```

```

        books[book] = False
        print("Book borrowed successfully.")
    else:
        print("Book is already borrowed.")
    else:
        print("Book not found.")

elif choice == "3":
    book = input("Enter book name: ").lower()

    if book in books:
        if not books[book]:
            books[book] = True
            print("Book returned successfully.")
        else:
            print("This book is already available.")
    else:
        print("Book not found.")

elif choice == "4":
    print("\nAvailable Books:")

    for book in books:
        if books[book]:
            print("-", book)

elif choice == "5":
    print("Thank you for using Library Management Assistant.")
    break

else:
    print("Invalid choice.")

```

Sample Output

===== Library Management Assistant =====

1. Search Book
2. Borrow Book
3. Return Book
4. Show Available Books
5. Exit

Enter your choice: 1
Enter book name: Python

python is available.

Enter your choice: 2

Enter book name: Python

Book borrowed successfully.

Enter your choice: 4

Available Books:

- java
- machine learning
- data science

Conclusion

The **Library Management Assistant** makes library operations easier by providing book search, borrowing, returning, and availability-checking features. It is a simple example of how an assistant can automate basic library tasks.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical / Concept	30%	All key concepts (AI, ML, DL, GenAI, LLMs) correctly identified with clear hierarchical relationships	Most concepts correct; minor gaps in relationships	Several concepts missing or misclassified	Major conceptual errors; map incomplete
Application	25%	Real-world examples linked accurately to each concept	Examples mostly relevant	Few or generic examples	No real-world linkage shown
Quality	20%	Logical, well-organized structure with clear connectors	Mostly organized; minor structural issues	Some disorganization affecting clarity	Poorly structured, hard to follow
Presentation	15%	Visually clear, creative, easy to interpret	Neat and readable	Cluttered or inconsistent formatting	Untidy, difficult to interpret
Documentation / Communication	10%	Concise labels/notes that add clarity	Adequate labeling	Minimal or unclear labeling	No supporting labels or explanation

Marks Evaluation:

Question No.	Technical / Concept(30%)	Application(25%)	Quality (20%)	Presentation (15%)	Documentation / Communication (10%)	Total Marks (Each – 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____